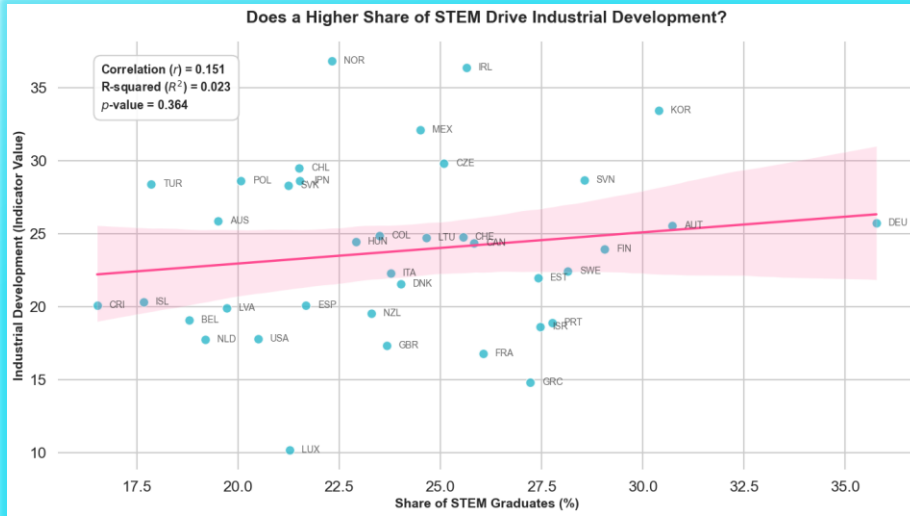


Data Visualization Pitch

Student Individual Assignment



Talents,
Industry, and
Richness: How
are they related?

Student name: Hao Li

Student ID: 947013

Student e-mail address: l.hao1@campus.unimib.it

Research questions

Purchasing power, share of STEM students, Earnings, Tax, the development of industry of a country, all these concepts are concerned by people everyday, **how are these concepts related to each other?** This research is aimed at showing people intuitions about those concepts.

- **Are the share of STEM students, or their salary level, related to the development of the industry of a country?**

Will the development of industry get improvement if there are more STEM students? Does the industry level imply a higher salary premium?

- **Is there a relation between purchasing power and the share of STEM students? And What about purchasing power after tax, and salary premium?**

It is because the country enriched by industry, or people are not willing to study STEM when they're too rich? It is average purchasing power, or purchaing power after tax, that is affecting the share of STEM students? Or just salary premium affects it?

- **Does the taxing rate itself affect the development of industry of a country?**

Will tax imporve or foster the development of industry?

About Data

- **Data Searching:** My initial intention was “**story-driven**”, so I firstly decided a list of data needed, and then searched on Google, but soonly I found it was very difficult to collect just right data, after several cycle of revising and collecting, I finally collected enough data on **WorldBank and OECD**.
- **Dataset Used:** Since there are too many of them, I put the data scoures in the [Appendix](#) page.
- **About the License:** The data are all share under License of **(CC-BY 4.0)**, claimed by World Bank and OECD. This mean I can copy, modify and distribute data in any format for any purpose, including commercial use, and of course to make derivative work, as long as I claim the data source.
- **Quality and Completeness:** Most of situations, the data are of high quality, but there are still many gaps and obviously outliers. I have to delete many rows, and for some missing values, I used **forward fill** and **linear regression** to fill a proper value. For obviously abnormal values (taxing rate of Colombia is zero), I **manually vertificate** the data from other source, shockingly found it is true.
- **Data Cleaning:** Most of cleaning work are deleting rows or filling missing values, which has been stated above.

Methodology

- **Data Collection:** I **manually downloaded** the data from multiple sources, there's no need of APIs or scrapers, since luckily both WordBank and OECD database provide data explorer to select data easily.
- **Data Processing & Cleaning:** For the reason of reproducibility, all the operations on dataset are made by Python, I developed Python in **Pycharm** and **Jupyter notebook**. I also used packages like **pandas**, **numpy** to process the data. and **matplotlib** to generate the charts and figures.
- **Data Transformation:** I merged different source of data to form a wide table, which is mainly performed by **double-key merging**, I used left-merging to add new blocks gradually, and finally formed a big wide table. Finally, I **aggregated** the wide table by Year through **time-weighted average**, this is aimed at improving data stability and not wasting information.
- **Use of AI Tools:** I used mainly **Gemini** to help with coding and brainstorm.
- **Analytical Techniques:** What methods did you use to derive insights? (e.g., descriptive statistics, trend analysis, comparative analysis, outlier detection)
- **Reproducibility:** I uploaded the project on Github:[https://github.com/Lihao-academic/Course Work Repo/tree/main/DataVisualizationProject](https://github.com/Lihao-academic/Course_Work Repo/tree/main/DataVisualizationProject)

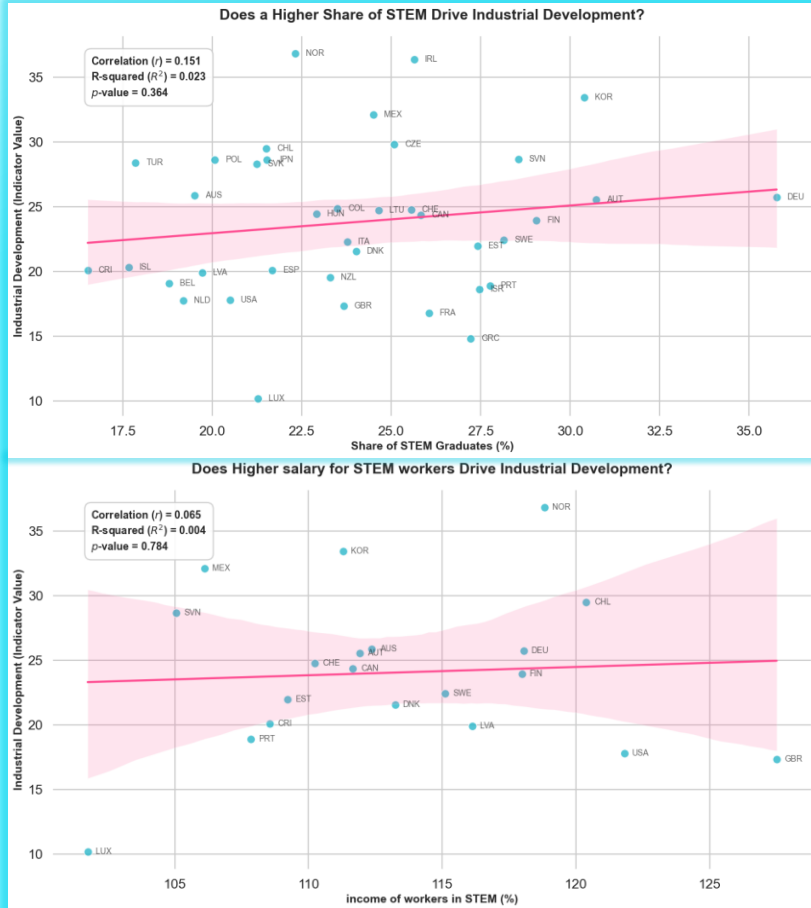
Concepts

- **PPP:** A metric that measures the true wealth of a country's citizens by adjusting for different price levels and exchange rates, allowing for a fair comparison of purchasing power across nations.
- **Taxing Wedges:** The ratio of total taxes and social security contributions paid by an employee compared to their total labor cost. Essentially, it is the slice of the salary pie taken by the government.
- **Industry value added, percentage of GDP:** A macroeconomic indicator showing the share of a country's total economic output that comes from industrial sectors (like manufacturing, mining, and construction) rather than services or finance.
- **STEM:** An acronym for the core technical and scientific disciplines that drive a nation's high-tech research, innovation, and industrial workforce.
- **Income of Workers in STEM (%):** The income premium or relative salary of STEM professionals compared to the national average salary. For instance, a value of 118% means STEM workers earn 18% more than the average worker.

Examples of the Final Combined Table

	Country Code	Country Name	PPP	Taxing Wedges	PPP after Tax	Share of STEM	income of workers in STEM (%)	Indicator Value
0	AUS	Australia	62622.728485	28.384989	44812.535669	19.500105	112.366610	25.864653
1	AUT	Austria	66664.760899	47.290181	35160.736265	30.727140	111.915770	25.562898
2	BEL	Belgium	64419.755455	52.621903	30527.805909	18.785552	NaN	19.089245
3	CAN	Canada	58285.109460	31.625546	39831.483622	25.825538	111.658737	24.365074
4	CHE	Switzerland	85606.510503	22.804733	66056.218599	25.554405	110.233687	24.762840
5	CHL	Chile	30196.007097	7.008208	28080.195776	21.510217	120.385898	29.474144
6	COL	Colombia	18939.137720	0.000000	18939.137720	23.494661	NaN	24.854569
7	CRI	Costa Rica	25921.884999	27.443611	18800.648119	16.519459	108.551503	20.109412
8	CZE	Czechia	50350.443456	41.497341	29530.656175	25.075742	NaN	29.812671
9	DEU	Germany	65624.199511	48.400769	33900.093544	35.769033	118.044703	25.752383
10	DNK	Denmark	71568.565631	35.887197	45875.993980	24.015019	113.243324	21.564063
11	ESP	Spain	49012.601017	40.204321	29263.995203	21.677391	NaN	20.100783
12	EST	Estonia	44886.181168	38.852170	27392.424624	27.416526	109.214038	21.962058
13	FIN	Finland	58528.113593	42.736605	33512.155264	29.059427	117.976396	23.932238
14	FRA	France	55267.134202	47.027585	29279.988297	26.050745	NaN	16.806594

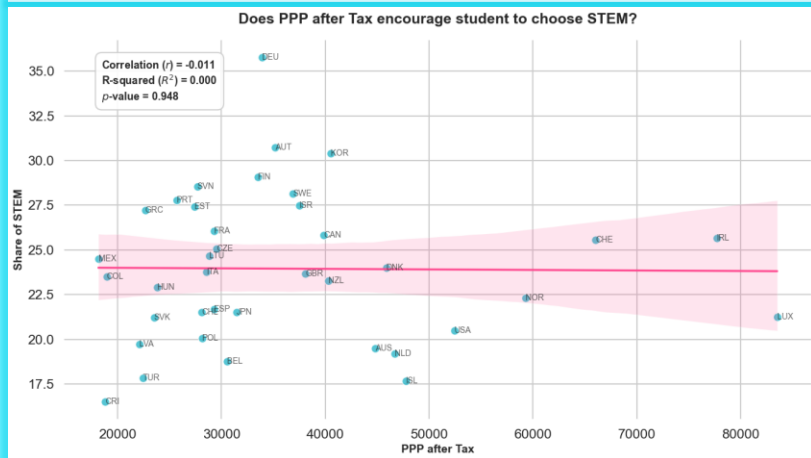
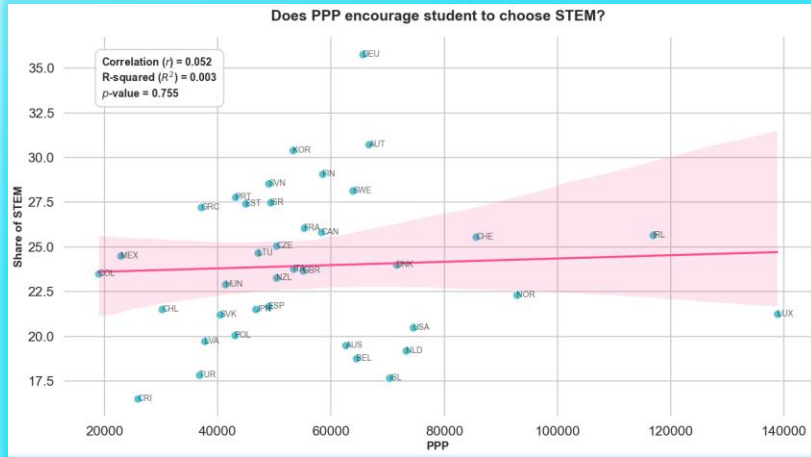
Question 1



However, the results were quite different from what we expected. In reality, there was little correlation between these variables. We can see that "Share of STEM grades" has a weak relationship with the industry indicator, although their p-values, correlations, and R-squared are not impressive. However, as we will see later, this is already the strongest correlation. Furthermore, in economics, it's inherently difficult to find clear correlations, so we reluctantly conclude that **a weak correlation** does exist here.

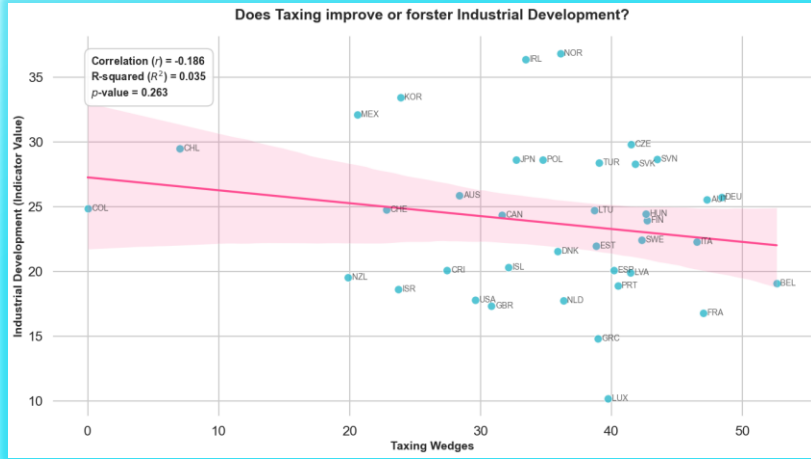
Conversely, their **salaries are almost unrelated to industrial development**, and industrial development does not necessarily mean a higher STEM salary premium.

Question 2



The second question we studied was about purchasing power parity and the proportion of STEM students in a country. However, we still found that there was almost no correlation. In other words, **people's average wealth level has almost no relation to their willingness to choose STEM fields, at least in OECD countries.**

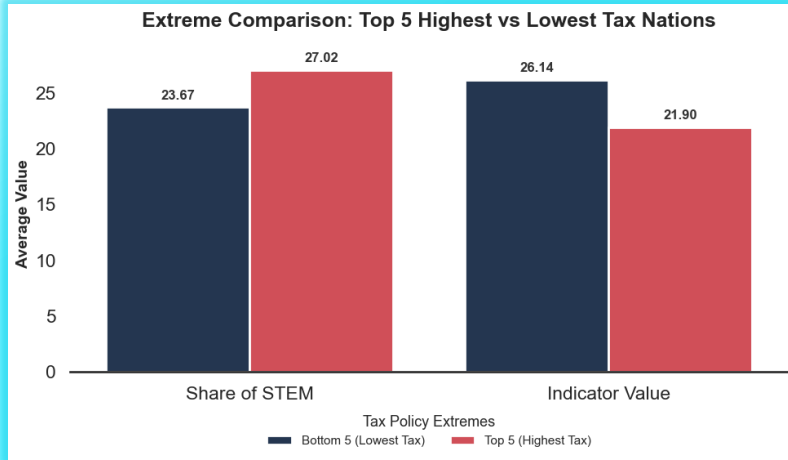
Question 3



Does the taxing rate itself affect the development of industry of a country? Will tax improve or foster the development of industry?

In the third question, we intend to study whether national taxes on workers affect the overall industrial development of the country. The results do show a slight negative correlation, and even the p-value is relatively low. We might say that **high taxes do have some negative impact on the overall industrial development** of the country.

Furthermore...



Since the previous correlations were relatively weak, we specifically compared the relationship between extreme high and low tax policies and the proportion of STEM students and industrial indicators. We found that under extreme conditions, this rule is still quite obvious. However, **high-tax countries do not affect students' enthusiasm**; on the contrary, they may encourage more people to choose STEM. **This may be because taxation plays a role in other aspects.**

On the other hand, **excessively high taxes can indeed hinder industrial development**, as we can clearly see in the graph.

Conclusions & Discussion

- Conclusions

- **The Wealth & Premium Paradox:** Highly industrialized or wealthier nations (higher PPP) do not naturally produce more STEM graduates. Furthermore, massive salary premiums (e.g., in the UK or Germany) do not guarantee a proportional surge in STEM students.**Beyond Macroeconomics:** The lack of strong linear correlations ($R^2 < 0.2$) is a major finding itself. It reveals that the "STEM Heaven" is not built on financial incentives alone; students' choices are likely shaped by complex socio-cultural factors and welfare structures rather than just money.

- Discussions

- **Lurking Variables:** Our model focuses on economic metrics, missing unmeasured variables that strongly influence education, such as immigration policies, gender gaps in tech, and cultural prestige.**Sample Size & Granularity:** The specific STEM income data limited our robust analysis to 21 OECD countries. Additionally, the broad definition of "Industry Value Added" might blur the differences between traditional heavy manufacturing and modern high-tech sectors.**Cross-Sectional Constraints:** By using a time-weighted average to handle sparse historical data, this cross-sectional study highlights structural associations but cannot definitively prove causality.

Appendix: Sources of the data used in the project

- **PPP:** [GDP, PPP \(current international \\$\) | Data](#)
- **Taxing Wages:** [OECD Data Explorer • Labour taxation – OECD comparative country indicators](#)
- **Share of STEM students:** [OECD Data Explorer • Distribution of enrolled students, new entrants and graduates by field of education](#)
- **Percentage of Income of STEM workers:** [OECD Data Explorer • Education and earnings \(EARNINGS\) \(full dataset\)](#)
- **Industry value added (% of GDP):** [Industry \(including construction\), value added \(% of GDP\) | Data](#)

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